

Now, add the root user to the *vboxusers* group:

```
$ sudo usermod -G vboxusers root
```

Since we are using VirtualBox 2.1, we don't even need to create a network bridge. All we do is plug in our VMs to eth1 (that's our client interface), and yippee, we are networked. eth1 has DHCP, so the virtual-servers are automatically configured. A word of caution: since we have a DNS system in our network, all our VMs should have good host names. I don't know what you named your domain, but suppose it's *officenetwork.local*, give your database server the name *db-server* and the mail server the name *mail*. So we can access them as *db-server.officenetwork.local* or *mail.officenetwork.local*.

Note that our architecture does not allow for accessing any of our VMs from outside (more precisely, eth0) our office. So a field officer trying to access his mail, will need to be able to VPN into our server. I'll set this up at a later stage.

Setting up our first VM: The Web server

First, some disappointing news... We will not set up our Web server now but just prepare it.

VirtualBox 2.1 can run 64-bit guests on a 32-bit host and vice-versa. So, since you have a Q6600 processor, you can seriously consider a 64-bit version of CentOS. And your processor supports hardware assisted virtualisation, so it's all the better.

First of all, head to *centos.org* and get yourself all the 5 CDs or the DVD (torrent download only) of the latest version of CentOS. As of now, CentOS 5.3 (corresponding to the RHEL 5.3 release) has not been released, so you'll have to settle for 5.2. However, CentOS 5.3 may come out any day now. You can upgrade as per your needs, easily.

I assume that you have downloaded the CentOS 5.2 x86_64 DVD and have saved it in */VMspace/ISOStore*.

So open up a terminal and type in the following:

```
$ sudo mkdir /VMspace/VDIStore
$ sudo VBoxManage createmv -name WebServer -register
$ sudo VBoxManage createvdi -filename \
/VMspace/VDIStore/WebServer.vdi -size 8192 -register
$ sudo VBoxManage modifyvm WebServer -hda \
/VMspace/VDIStore/WebServer.vdi
$ sudo VBoxManage modifyvm WebServer -dvd \
/VMspace/ISOStore/CentOS-5.2-x86_64-bin-DVD.iso
$ sudo VBoxManage modifyvm WebServer -memory 512M
$ sudo VBoxManage modifyvm WebServer -acpi on \
-ioapic on -vrdp on -vrdpport 3390
$ sudo VBoxManage modifyvm WebServer \
-ostype RedHat_64 # change if using 32bit
$ sudo VBoxManage modifyvm WebServer -pae on \
-hwvrtex on -accelerate3d on
```

Now, to configure some sophisticated networking. This is possible from the *VBoxManage* command line as well, but to better understand the procedure, let's do this from the GUI.

Launch the GUI, select *Web Server* from the VM list and then click on the networking hyperlink on the right. In the pop-up box, enable Adapter 1. Then, select its type to be 'Intel Pro/1000 T Server (82543GC)'. Choose 'Host Interface' in the *Attached To* field. Generate a random MAC address (do not change this in the future) and finally attach the host interface to eth1 (our internal network).

So NASA has just given us the Go-code for the launch! However, you soon abort the launch after noticing a fatal error—it's impossible to access the Web server from outside (eth0). Well, relax! What are reverse proxies for, anyway? By the way, give nginx a miss; we will set up a different configuration another day.

Fine-tune any settings as you see fit (if you need audio, you can set it to AC'97) and then drop down to the command line. Go to your Windows machine and start up the Microsoft Remote Desktop Connection. On the address field, use the server FQDN (server.officenetwork.local, for example) and append a "3390" to it. Expand the options button, click on the experience tab and select "LAN (10 Mbps or higher)". Don't connect as yet.

Type in the following command at the server console:

```
$ sudo VBoxManage startvm WebServer -type vrdp
```

You should get a pop-up window running *VBoxHeadless* with the VM. Immediately, connect the RDC on the Windows machine. Voila!! You have the Remote Access part of VirtualBox running as expected. Don't boot up now, abort the GRUB countdown and issue the following:

```
$ sudo VBoxManage stopvm WebServer
```

Launch up the GUI, and continue installing CentOS 5.2. Why I told you not to set-up remotely is because X behaves radically if you don't have the VirtualBox Guest Additions installed.

At this point don't set up any server tools at all—just set up a basic X desktop environment, preferably GNOME (because it is lighter). Just have one partition (and a swap-space), no LVM (the KiSS strategy). Give it a descriptive hostname ('web' would do fine) and set it up to configure its network over DHCP. Oh, and disable SELinux and the firewall in the text-mode part that comes after Anaconda's finished.

Install all the VirtualBox Guest Additions now. RDP into the machine (no problem if the GUI is running as well; anyone sitting at the GUI will think that a ghost has possessed the VM) and see if it works well. It should.